

SimDash: Unity Arduino Car Simulator

HTI.320 Group 7 Final Presentation

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Introduction

What is SimDash?

- Interactive driving simulator combining hardware and software
- Real-time feedback through physical components
- Unity-based 3D environment with realistic physics
- Arduino-controlled peripherals for immersive experience

Core Concept: Bridge the gap between virtual simulation and tangible interaction

Team Roles & Responsibilities

Software (Unity):

- 3D driving environment
- Realistic physics engine
- Weather simulation (rain)
- Random event scenarios
- Serial communication interface
- Speed monitoring & warnings

Hardware (Arduino):

- Arduino Uno microcontroller
- 3-digit 7-segment display (5641AS)
- 5V fan motor with PWM control
- Steering wheel input device
- USB serial connection

Bidirectional data flow: Unity ↔ Serial ↔ Arduino

Prototype Demo

Key Demo Features

1. Real-time Speed Display

- Speed from Unity transmitted via serial
- Arduino parses and displays on 7-segment display
- Updates dynamically as car accelerates/decelerates

2. Wind Flow Simulation

- Fan speed synchronized with car speed
- Adds immersion in virtual environment

3. Interactive Driving Environment

- Steering wheel and pedals
- Simple physics-based movement and collision
- Weather effects impact visual experience

Proposal vs. Final Product

What Changed?

Removed Proposed Features:

- Heart rate sensor
 - Delayed delivery
 - Time constraints
- ECG helmet integration
 - Complex implementation
 - Not available for student use
 - Scope too large
- Automatic stress-based control

Enhanced Instead:

- Polished driving environment
- Improved weather effects
- Better audio integration
- More stable hardware setup
- Refined physics simulation
- Random scenario system

Result: More focused, achievable prototype that still demonstrates core concepts

Development Process

Tools & Workflow:

▪ GitHub Repository

- Version control for code
- Branch-based development
- Main branch stability
- Merge reviews

▪ Miro Board

- Visual planning
- Task tracking
- Milestone setting

▪ Documentation

- README.md (setup guide)
- Circuit diagrams
- Code comments
- Final report

▪ Testing Approach

- Standalone Arduino tests
- Unity scene testing
- Integrated system validation

Repository: <https://github.com/3liasP/simdash>

Lessons Learned

Key Lessons:

- **Hardware Risks:** Component availability is important from the start
- **Integration Early:** Test hardware-software connection ASAP
- **Parallel Work:** Not so simple with Unity Scenes to avoid conflicts
- **Flexibility:** Be ready to pivot when obstacles arise

**Despite challenges, we delivered a functional prototype
that demonstrates our core concept successfully!**

Thank You!

Questions?

Repository:



Miro Board:

